

Solution Requirements (FR & NFR)

Date:	15 March 2026	Team ID:	PNT2022TMID124356
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Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Secure log-in, registration, and JWT token handling.	High
2	Authorization levels	Only authenticated users can view/modify their own loan accounts, analysis history, letters, and advisor chats.	High
3	External interfaces	Connection to external Google Gemini API to query settlement metrics and strategies.	High
4	Transactions processing	Ratios calculations (DTI, stress scores, disposable surplus) executed dynamically on database adjustments.	High
5	Reporting	Direct streaming and compile download of PDF summaries and Excel workbooks.	Medium
6	Business rules, etc.	Delinquent loans have priority flags. Calculations automatically check status categorizations.	High

Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Low-latency response for all calculation routes and graph queries.	Page loads < 200ms; AI calls < 2.5s
2	Scalability	Supports local virtualization and container orchestration.	Docker-Compose compliant
3	Security & Data Privacy	Encrypted credentials storage; JWT authorization scopes.	Random salt PBKDF2 HMAC SHA-256
4	Reliability	Uptime and local database write persistence.	99.9% local uptime availability
5	Usability	Glassmorphic UI with active route menus.	Light & Dark theme responsive layouts